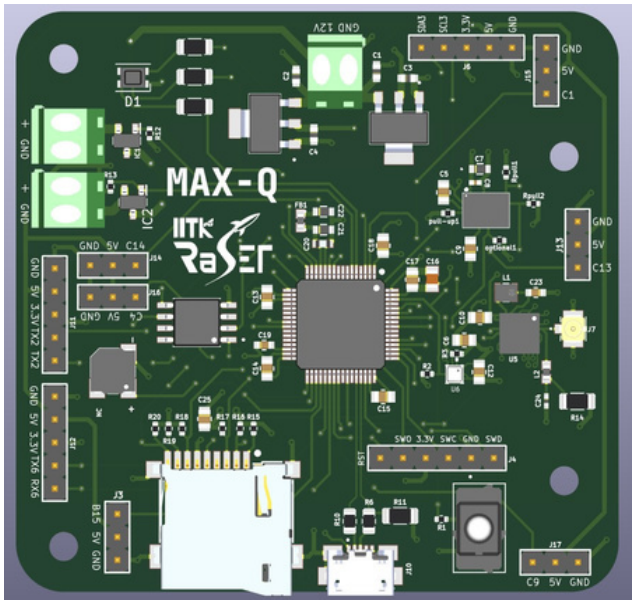

Avionics, Payload & Control (APnC)

General Instructions

Hey, upcoming rocketeers! Ever wondered how the rocket knows what it is doing, where it is and where it needs to go?
Welcome to **Avionics**, **Payload** and **Control** Subsystem!!!
Following is the material to get you started with the basics - it might look daunting but most of them are small videos so sit tight with a cup of coffee ☕ and enjoy.

How to (and not to) go about APnC:
Rocket Avionics - Building Lumineer

Avionics

ELECTRONICS AND FLIGHT COMPUTERS

- Arduino 101- Crash Course w/ Mark Rober
- Easy & Powerful Arduino Alternative? STM32 Beginner's Guide
- I2C | Arduino Docs
- Electronic Basics #19: I2C and how to use it
- Arduino & Serial Peripheral Interface (SPI)
- Electronic Basics #36: SPI and how to use it
- UART | Arduino Docs
- Understanding UART
- Flight Computer Outline - Landing Model Rockets Ep. 3

TELEMETRY

- Design of a Radio Telemetry System
- Sample Rocketry Telemetry System - Good Read!

ANTENNA

- Antenna Basics
- What is Antenna Impedance? (Explained)
- Also, check out the link budget calculation and try to play around with the antenna. Hope you find something new.😊

- We use 4nec2 software simulation of antennas.
- Dipole Antenna design using 4 NEC2 Simulator
- It stores files in the form of .nec file format.
- For practice, u can try out making QFH antenna in 4nec2 simulator (Hint: write python code to generate .nec file for this)
- Helix Antenna | Making A L-Band Helix Antenna For Decodi…

Payload & ML

What is a Payload?

Rockets and Payloads - A Beginner's Guide to Rocket Vocabulary

a) Payload is essentially the experimentation module on-board a rocket. In full-scale rocketry, the payloads are generally the satellites or human space capsules.
b) For Model Rocketry,the payload can be a CANSAT,separable or not, which can carry different technologies on board. These might be atmospheric sensors, biological experiments, anti-gravity experiments and much more.

Our current work in the “Payload” subsystem also involves using Machine Learning to predict future time coordinates in space,as well as orientation. Below videos give some insight to the techniques used onboard our payloads,in addition to some Machine Learning architectures frequently utilized for this.

- What is Time Series Analysis?
- Recurrent Neural Networks (RNNs), Clearly Explained!!!
- Time Series forecasting with XGBoost

You may think of innovative technologies/ideas/experiments to integrate into our payloads. We might ask you for some suggestions in the recruitment test 😉

Control

We have now stepped into the domains of active control of our rockets and CANSATs where our current work includes roll control of rocket & altitude (orientation) control of CANSATs.

Apart from these projects, we are also steering ourselves towards guided rocket recovery (quite literally!) and real-time apogee regulation.

- How Chandrayaan 3 landed? (Fun Watch)
- Introduction to System Stability and Control
- How To Build a Thrust Vectored Model Rocket
- An Attempt at Roll Control
- Building a Supersonic Rocket Guidance System
- Intro to PID
- PID Examples
- Control Theory Basics

For any doubts, please feel free to contact

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